

Ryu Hemingway

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PROFILE

MSCS Align graduate student at Northeastern University building AI/ML systems and software engineering expertise, with a prior background in clinical medicine and behavioral research that brings rigor, structured analytical reasoning, and high-stakes documentation experience to technical work.

EDUCATION

Northeastern University — Silicon Valley, CA

September 2025 – Expected May 2027

Master of Science in Computer Science (Align Program)

- Coursework: Fundamentals of Computer Science (CS 5001), Discrete Structures (CS 5002), Object-Oriented Design (CS 5004), Data Structures and Algorithms (CS 5008), Algorithms (CS 5800), Foundations of Artificial Intelligence (CS 5100)

St. George's University School of Medicine — Grenada

September 2023 – June 2024

Completed first year of Doctor of Medicine program before pivoting to Computer Science

The Ohio State University — Columbus, OH

August 2017 – May 2021

Bachelor of Arts in Psychology

- Dean's List, Fall 2020

TECHNICAL SKILLS

Programming Languages: Python, Swift, JavaScript, SQL, HTML/CSS

Frameworks & Tools: React, SwiftUI, SQLite, AWS (Lambda, DynamoDB), Git, GitHub, Visual Studio Code, Command Line/Terminal, EMR Systems

Concepts: Object-Oriented Programming, Data Structures, Algorithms, Systems Thinking

HONORS & CERTIFICATIONS

- Certified Medical Scribe — ScribeAmerica, Spring 2021
- Dean's List — The Ohio State University, Fall 2020

PROJECTS

Ghost — Native macOS AI Assistant

2026

- Built and shipped a native macOS menu-bar AI assistant in Swift and SwiftUI (58,000+ lines across 136 source files), routing prompts to local LM Studio and Ollama models or to hosted Claude, Gemini, and DeepSeek APIs from a single interface
- Designed a capability harness of 75+ app-owned tools (file and document generation, format conversion, Calendar and Reminders, OCR, shell) that verifies permissions and path boundaries, executes the action, and returns the real result instead of trusting the model's claim
- Implemented a local retrieval-augmented generation pipeline on SQLite FTS5 with on-device semantic recall, indexing 30 document and source-code formats and returning cited, source-backed answers
- Enforced a privacy invariant that keeps Messages, Notes, and Mail content on-device and out of every cloud provider request, and wrote 450+ unit tests covering routing, tool execution, and egress policy
- Distributed the app as a signed, notarized release with Sparkle auto-updates, an AWS Lambda and DynamoDB licensing service, and a CloudFront and S3 landing page

Personal Portfolio Website

2025

- Built a responsive single-page portfolio site using HTML, CSS, and JavaScript, deployed via GitHub Pages
- Designed terminal-inspired dark theme with custom typography and teal/purple accent system

[In progress: fine-tuning evaluation project]

RELEVANT EXPERIENCE

UCSF Benioff Children's Hospital Oakland — Oakland, CA

August 2021 – May 2022

Medical Scribe, Pediatric Orthopedics

- Documented 800+ hours of complex diagnostic data across concurrent patient cases, maintaining accuracy under time pressure in a high-volume Electronic Medical Records system
- Reduced transcription errors by developing and refining documentation protocols adopted across the scribe team
- Trained 3 incoming scribes on specialty-specific workflows and process optimization, accelerating ramp-up time

LEADERSHIP & VOLUNTEER EXPERIENCE

American Red Cross — Disaster Action Team, Columbus, OH

Aug 2019 – Feb 2022

Disaster Relief Volunteer

- Coordinated emergency response logistics across multiple deployments, managing rapid crisis assessment
- Built case tracking systems for victim resource allocation, improving data accessibility for field teams

Operation Smile — Bangkok, Thailand

Feb 2015 – Apr 2015

Event Organizer

- Led cross-functional fundraising and data-driven outreach, raising \$30,000+ for cleft palate surgeries and driving event attendance to 5,000+ participants